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PEDIATRICS • REVIEW

Exploring the physiological mechanisms of sleep's influence on athletic performance and recovery: a narrative review

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Abstract

Purpose This narrative review explores the complex relationship between sleep, athletic performance, and recovery, emphasizing the physiological mechanisms at play.

Methods This review included papers reporting physiological mechanisms linking sleep, athletic performance, and recovery. Literature was identified through targeted searches in relevant databases and bibliographies of referenced scientific articles. Retrieved studies were analyzed thematically according to major domains, enabling integration of foundational mechanisms with applied evidence to inform practice and future research.

Results Slow-wave sleep (SWS) plays an important role in physical recovery, with its increase during the night being linked to rises in metabolic stress and growth hormone production. When training volume and/or intensity increase, a corresponding increase in SWS is usually observed, but a fine line may exist between promotion and disruption of sleep (i.e., overtraining). Acute sleep deprivation or restriction may influence sports performance differently, depending on the sport. In sports requiring complex skills or high cognitive demands, impairments may be more pronounced. Among the main strategies, napping has received the greatest attention, as it serves as a supplement to nighttime sleep and is associated with improvements in fatigue and sleepiness perception and physical and cognitive performance. Sleep extension has been studied less, but existing evidence shows improvements in cognitive and physical performance, as well as sport-specific skill execution.

Conclusions Despite the growing body of research on athletes' sleep patterns and their impact on performance, there is limited focus on the underlying mechanisms and practical benefits of sleep.

Keywords Sleep · Fatigue · Recovery · Athletes · Mechanisms · Muscle damage

Introduction

Sleep is a basic physiological need defined as a recurring behavioral state of reduced movement and responsiveness, altered brain activity, and consciousness of the surroundings practically suspended. It is an important component of human life, occupying between 20% and 40% of most individuals' day. Some general benefits of sleep include improving memory recall, regulation of glucose metabolism, and reducing mental fatigue while also playing an essential role in tissue repair, synaptic homeostasis, and immune-inflammatory responses [1–3]. Its therapeutic benefits include the removal of toxic waste byproducts in the brain, which have accumulated throughout the day [2]. Adequate sleep consists of ample duration, high efficiency (i.e., the percentage of time in bed that was spent asleep), and a proper routine [4–6]. With adequate sleep, individuals should feel refreshed and maintain alertness during the awake period

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[7]. For the adult population, the recommended sleep duration consists of seven to nine hours per night, as per current recommendations, to promote general health [8]. At the same time, lower durations can be considered suboptimal and likely to compromise health, well-being, and physical performance [8].

Sleep research in athletes has grown substantially over the years, with 82% of the studies being published between 2011 and 2019. In 2019 alone, 57 peer-reviewed research papers (18%) were published [9]. This substantial body of research has yielded valuable insights into the challenges and issues surrounding athletes' sleep characteristics, highlighting potential areas for improvement. However, despite portraying sleep as a recovery strategy, most original studies, which are primarily observational and descriptive, have focused on the relationship between sleep and sports participation-related factors (i.e., training loads, training, and match schedules, travelling) and how one can negatively impact the other (e.g., the impact of poor sleep on sports performance or the effects of sports-related outcomes on sleep quality). On the other hand, the benefits of sleep on physical performance and recovery, as well as the physiological mechanisms in play, have received less attention. It must be acknowledged that sleep is a very complex process, which makes it particularly challenging to isolate and accurately estimate its effects on any specific outcome, such as muscle injury, recovery, or sports performance. Thus, this brief review aims to synthesize the scientific literature related to the underlying mechanisms through which sleep affects athletic performance and recovery, as well as evidence from sleep intervention studies, to understand better what happens when we sleep and how it translates into practice.

Underlying mechanisms responsible for the relationship between sleep and physical performance and recovery

Sleep is a complex physiological process linked to the function of every organ system in our body, and a disruption in this process has important consequences for health and well-being [10]. Despite the advancements in clinical and basic sleep research and the widespread health consequences of sleep disturbances, much uncertainty remains [10].

Sleep is essentially divided into two phases: Rapid eye-movement (REM) and non-REM sleep. REM sleep has an important role in memory and learning consolidation, brain stimulation, and localized recovery process [11]. It is associated with dreaming, and electroencephalography analysis has revealed similarities with the awake period [12]. REM sleep is also extremely important in promoting growth and development of the brain and nervous system [11], which is supported by the prevalence of REM sleep in the early years of human life [13]. Newborns sleep for approximately two thirds of the day, and REM sleep occupies half of the total sleep time. It has also been associated with a decrease in norepinephrine levels, a chemical responsible for stress and emotions [14], thus allowing some "mental recovery". On the other hand, non-REM (NREM) sleep is considered important for somatic function and the replenishment of cerebral glycogen stores that are steadily depleted during waking [15]. Specifically, slow-wave sleep (SWS) plays a vital role in athletes' recovery, considering that SWS increases following a rise in metabolic stress [15]. Sleep, particularly SWS, is accompanied by an increase in growth hormone (GH), which may provide an ideal environment for anabolic function [16]. When GH release occurs [17], NREM sleep is physiologically stable compared to wakefulness and REM sleep [18], being important for tissue reconstitution and physical recovery.

Promotion of sleep through exercise and sleep adaptive function

While sleep has a big role in physical recovery, physical exercise also promotes sleep. For instance, physical exercise and demanding training sessions can increase the proportion of SWS. To explore the relationship between sleep and recovery following physical fatigue, the impact of several types of physical exercise was originally explored in non-athlete participants [19]; findings included increases in SWS during nighttime sleep that were related to the amount of physical fatigue, while REM sleep was reduced. Other studies then confirmed these results, with increases in stages 2 and 3 and decreases in REM sleep when the amount of running performed was manipulated [15, 20]. Of course, there may also be a delicate balance between the extent to which increased training loads promote SWS and its restorative function and the potential for excessively high training loads to disrupt sleep. This scenario is perhaps more plausible for athletes than for active individuals [17, 21].

In athletes, an increase in training load was associated with an increase in SWS and improved sleep continuity [22]. In swimmers, the proportion of SWS increased at the beginning of the season and during an overload period compared to a taper period [23]. In an ultra-endurance cyclist who engaged in a multiday ultra-endurance event, a progressive day-to-day increase in SWS was observed [24]. Evidence of increased SWS following heavy training sessions or competition has also been reported in some reviews [25, 26]. Interestingly, in extreme duration endurance events, such as ultra-marathons, it is consistently shown that sleep continuity is disturbed and sleep architecture altered [15, 27]. Here,

N3 sleep stage is favored, rather than REM sleep, indicating a key role of SWS on recovery after extreme endurance events.

Nevertheless, this should not be surprising. Besides its known importance for the restorative function mentioned above, such as anabolic hormone release and cleaning of cerebral metabolites accumulated during waking [3], SWS, along with sleep onset latency, serves as the primary driving mechanism in the homeostatic sleep process and is a reliable indicator of sleep needs [28]. Thus, considering that heavy/intense exercise might induce accentuated muscle damage, delayed-onset muscle soreness, and inflammation response [29], we can hypothesize that the daily accumulation of training load can maximize sleep pressure and/or increase the physiological restorative demands, promoting SWS. This supports the adaptive nature of sleep to physical activity [3, 26, 30].

Additionally, evidence suggests that adenosine, a neuromodulator accumulating in the brain during wakefulness, plays a central role in linking metabolic activity with sleep homeostasis. Increased adenosine concentrations in response to elevated energy turnover and the accumulation of exercise related metabolites may act as a signal for enhanced sleep pressure and slow-wave activity [31]. Studies with rodent models have demonstrated that adenosine receptor activation, especially in the basal forebrain and cortex promotes SWS and have a role in recovery from exercise-induced metabolic stress through AMPK phosphorylation and subsequent regulation of cellular energy balance [32, 33]. This interaction between metabolic activity and sleep highlights how energy metabolism and adenosine signaling may underlie the adaptive increase in SWS observed with higher training loads.

Hormonal changes affected by sleep duration and their relationship with muscle recovery

Since sleep and the circadian system play an important role in regulating the hormonal and immune systems, sleep duration is thought to play a role in muscle repair. In athletes, sleep deprivation has been shown to increase basal and post-exercise serum levels of skeletal muscle enzymes or proteins which are considered indicators of muscle damage, such as CK, aspartate aminotransferase, alanine aminotransferase, C-reactive protein (CRP), and myoglobin [34]. In animal models, this catabolic environment has been associated with muscle impairment (i.e., increase of muscle TNF-α levels), which might slow the regeneration process after muscle damage or injury [35].

During sleep, several hormones associated with protein synthesis are released [36, 37]. Muscle health is strongly influenced by hormonal secretion, where testosterone, GH, and IGF-1 are known to increase the activity of protein synthesis [38]. Testosterone levels, for example, are related to the first episode of REM sleep at night and have anabolic effects [39], which lead to protein synthesis [40]. However, sleep restriction may lead to significant reductions in testosterone levels, as a 10–15% reduction was observed after just one week of sleep restriction (5 h per night) in young adults [41]. Insufficient sleep has also been associated with increased adenosine, a neuromodulator with a general inhibitory effect on neural activity, inhibiting neural drive [42]. The response of blood cortisol concentrations to sleep deprivation are similar to those with sleep restriction, but findings among the two are inconclusive and uncertain [43–45]. If increased cortisol concentrations do happen after sleep restriction or deprivation [45], this could lead to increased muscle catabolism and a reduction in protein synthesis [46]. This would strengthen the restorative theory that sleep is required for muscle recovery [47]. However, this hypothesis requires further research. Lastly, sleep disorders and increased sleep duration variability are shown to be linked with higher levels of CRP and IL-6 [48]. This is important since IL-6 plays a key role in hypertrophic signaling and muscle restoration [49].

Sleep and physical performance

Injury risk and recovery from injury

Sleep may also be associated with the incidence of injuries, although current evidence does not seem to support poor sleep as an independent risk factor for increased injury risk [50]. It is possible, however, that sleep acts as a moderator of other potentially modifiable risk factors [50]. For a more in-depth review of the associations between sleep and the incidence of injuries, the reader is referred to the review by Dobrosielski et al. [50].

In this type of study, sleep is typically assessed using subjective questionnaires, which may not always be validated for use with athletes, rather than objective monitoring. Additionally, authors often employ univariate analyses instead of more complex multivariate models [50]. Bittencourt et al. [51] suggest examining sports injury from a complex systems approach, considering the importance of multidirectional interactions between factors and their potential influence on risk identification and subsequent injury occurrence.

Based on available literature, one can theorize about the relationship between sleep physiology and injury incidence. For example, sleep deprivation may trigger intermediary mechanisms that contribute to the occurrence of injuries. Testosterone and GH are metabolic hormones involved in

protein synthesis; however, one week of sleep restriction leads to a significant reduction in testosterone levels in men [41]. Additionally, plasma concentrations of GH peak within 120 min of sleep onset, but when sleep is disrupted, GH levels are reduced [52]. On the other hand, Cortisol is released in higher concentrations following one day of poor sleep [53]. All these changes in hormonal secretion patterns with sleep disruption may impair skeletal muscle integrity and body recovery from exercise, which may lead to injury. Although acute increases in anabolic hormones, such as Testosterone, have no effect on muscle protein synthesis [54], chronic low Testosterone levels, alongside increases in blood Cortisol may be linked to insufficient muscle restoration and promote muscle degradation [55, 56].

Between-sports comparisons on the effects of sleep on performance

Evidence regarding sports-specific outcomes is limited. In swimmers, limiting athletes to 2.5 h of sleep per night during 3 consecutive nights resulted in no impairment in the 50 and 400 m swim times [57]. It has been suggested that on shorter, maximal events, such as sprinting, sleep debt may produce smaller detrimental effects on performance when compared to prolonged submaximal efforts [58]. This may be due to less impact of the increased perceived exertion or muscle glycogen deficits in shorter events [59, 60]. On the other hand, sports involving high cognitive demanding tasks may be more influenced by short sleep durations. Dart throw performance (accuracy and consistency) was impaired following 3–4 h of sleep reduction [61]. Tennis serving accuracy was impaired after reduced sleep in collegiate players [62]. Another study showed that sleep restriction impacts tennis-specific tasks (service and crosscourt shots) but not athletic performance (i.e., repeated sprint ability) [63]. Recently, a universal effect of sleep disruption on cognitive performance has been proposed [64] due to arousal and attention in a sleep-disrupted state [65]. The neuroanatomical mechanisms behind this state are intricately complex [66]. It appears that when the quality and quantity of sleep is reduced, the largest decreases in cerebral metabolism happen in the thalamus, cerebellum, and prefrontal, posterior parietal, and temporal cortices [66, 67]. The reduced metabolic rates within these regions of the brain have been correlated with decreased cognitive performance [68].

Although this topic lacks investigation, athletes from sports with high cognitive demanding tasks thus may see their performance more negatively impacted by short sleep durations. However, it is possible that the impairment of cognitive function may also negatively affect athletes' performance in sports that include highly technical actions (i.e., batting in baseball). Decreased self-perceived capacity or increased perceived exertion may also decrease time to fatigue in sports that require intermittent and repeated bouts of exercise and increase decision-making errors in sports that are played over prolonged periods of time. In addition, very little evidence is available regarding the impacts of habitual short sleep durations on sports performance, which is very common among the athletic population [69].

Practical benefits of sleep interventions on recovery and performance

Since sleep is considered a recovery strategy, it would be reasonable to link it with markers of recovery. For example, cold water immersion, one of the most known and utilized recovery strategies [70, 71], has been associated with performance markers such as the countermovement jump, sprint and accumulated sprint times, perceived muscle soreness and fatigue, and biochemical markers such as CK, lactate and pH [72]. Nevertheless, it should be recognized that contrary to other recovery strategies, sleep modifications are not entirely dependent on the athlete's will [73], making it, in part, harder to explore and apply in practical scenarios such as high performance sports.

Nap

Several studies have been published recently, presenting strong evidence that napping may aid recovery between multiple training sessions (i.e., morning and afternoon training) while also representing a supplement to night-time sleep or reducing daytime sleepiness, particularly following accumulated sleep debt [74].

As for the effects of napping on the perception of effort and fatigue, most studies conclude that when athletes take a daytime nap, they exhibit a lower perception of fatigue [75, 76], feel less tired and stressed [76, 77], less angry or depressed [75, 78], and report better mood states [75]. These effects are likely due to the fact that napping, even with a short episode of SWS, can reduce the homeostatic pressure of sleep [79]. On that, napping later in the day reduces SWS proportion in the initial part of the subsequent nocturnal sleep [80], a consequence of SWS's homeostatic regulation [81]. Several studies have reported lower perceived exertion during aerobic or anaerobic exercise after a daytime nap [75, 78, 82–84]. However, some other studies have failed to find an impact of napping on post-exercise perceived exertion [85, 86]. These conflicting results might be related to differences in the study protocols, especially controlling the night(s) before the daytime nap. Allowing normal sleep on the night before [82] or applying a 5-night severe sleep deprivation condition [85] will severely impact the results.

Also, the nap duration and time between the nap and testing can lead to different results. With several reviews on napping strategies being published, the reader is referred to the reviews of Mesas et al. [87], Lastella et al. [88], Botonis et al. [89], Boukhris et al. [90], and Boukhris et al. [91] for in-depth details on this strategy.

In summary, under well-rested conditions (>7 h of night sleep), daytime napping may improve physical and cognitive performance and reduce perceived fatigue [87, 89]. In this scenario, a daytime nap may provide the additional sleep duration required for sufficient recovery. Napping following a normal sleep night seems to contain mostly stage 2 NREM sleep and relatively low SWS [15, 84]. In fact, a shorter proportion of SWS may lead to less sleep inertia following the nap. On the other hand, for athletes experiencing short sleep durations, daytime napping may attenuate performance decrements induced by sleep loss. In this case, the proportion of SWS following a daytime nap is much greater, which could be due to the role of SWS in the recovery process. Most studies that found improvements in performance following a nap also noted that these increments were accompanied by reduced sleepiness/increased arousal and improved perception of recovery [76, 77, 85, 92].

Sleep extension

While many studies have examined the effects of less sleep on recovery and physical performance, very few studies have explored the exact opposite. The main benefits reported from extending sleep time are decreased sleepiness and a positive effect on cognitive function [93–96]. Considering that studies often report less than ideal sleep durations in athletes [30, 69] and that this specific group may benefit from sleeping more than what's currently recommended for healthy adults [93], sleep extension would be a solution. Like daytime napping, sleep extension can be important for individuals experiencing sleep difficulties or sleep deprivation over the past few days. However, the ratio of "sleep loss" to "sleep recovered" is not exactly a 1:1 ratio. This is, if an individual who usually sleeps 8 h a night is totally deprived of sleep for one night, he/she will not require 16 h of sleep to achieve full sleep recovery. This phenomenon suggests that either recovery sleep is more efficient than usual nighttime sleep (greater minute-by-minute recuperative value) and/or the rate at which sleep-furnished resources are used during wakefulness is reduced by prior sleep loss [97].

Most studies exploring the effects of sleep extension on recovery and physical performance have found similar results. Specifically, it can improve sprinting time [93], time trial and endurance performance [98, 99], psychomotor vigilance task [93, 98], mood [93], reaction time [95], and decrease sleepiness [93, 94]. It seems that cumulative sleep time affects performance by reducing the perceived exertion for a given exercise intensity. Sports-specific improvements include shooting accuracy in basketball [93], and serving accuracy in tennis [94], for example. In rugby athletes, it also decreased physiological stress (i.e., lower cortisol levels), but did not improve daytime sleepiness, immune function or sympathetic nervous system activity [95]. Of note, using an individualized sleep extension strategy based on each athlete's usual sleep duration rather than an arbitrary increase may be a better strategy for future studies.

In healthy young adults, sleep extension increased IGF-I levels [100], which have been shown to contribute to improved cognition [101, 102]. Thus, increased IGF-I levels may partially explain beneficial effects on attentional performance and alertness during total sleep deprivation and recovery [103]. Sleep extension also showed improvements in different aspects, such as time to exhaustion and perceived exertion post-exercise [104], and fatigue perception [96].

An important factor in sleep extension is the ability to extend the amount of sleep beyond athletes' normal sleep, which seems to be a psychophysiological feature [105]. Although there is evidence that individuals who are already sleeping 7–8 h per night are able to obtain up to 3.5 h of additional sleep, it is not this straightforward [106]. Anecdotal evidence shows that the ability to sleep for 10 or more hours is a distinctive feature of some individuals [107]. It is also important to consider and report the subjects' chronotype. This is important since individuals who perform physical exercise for their chronotype (i.e., evening-type training at night) will likely perform better at those times than at times opposite to their chronotype [108].

Chronotype

Chronotype itself may play a role in the recovery process as well. Individual chronotype reflects differences in the internal circadian phase that influence the timing of sleep–wake propensity, dim light melatonin onset (DLMO), core body temperature, and the circadian timing of hormone secretion [109]. Chronotype therefore not only determines behavioral preferences for training and competition times but also modifies the endocrine environment that supports recovery. For example, many anabolic and recovery-related hormones follow distinct circadian patterns: growth hormone and slow-wave sleep are closely linked to early-night NREM sleep, cortisol shows a robust morning peak, and testosterone exhibits nocturnal rises that are phase-dependent [110–112]. Misalignment between training/competition schedules and an athlete's endogenous circadian phase may blunt these hormone rhythms and alter sleep architecture

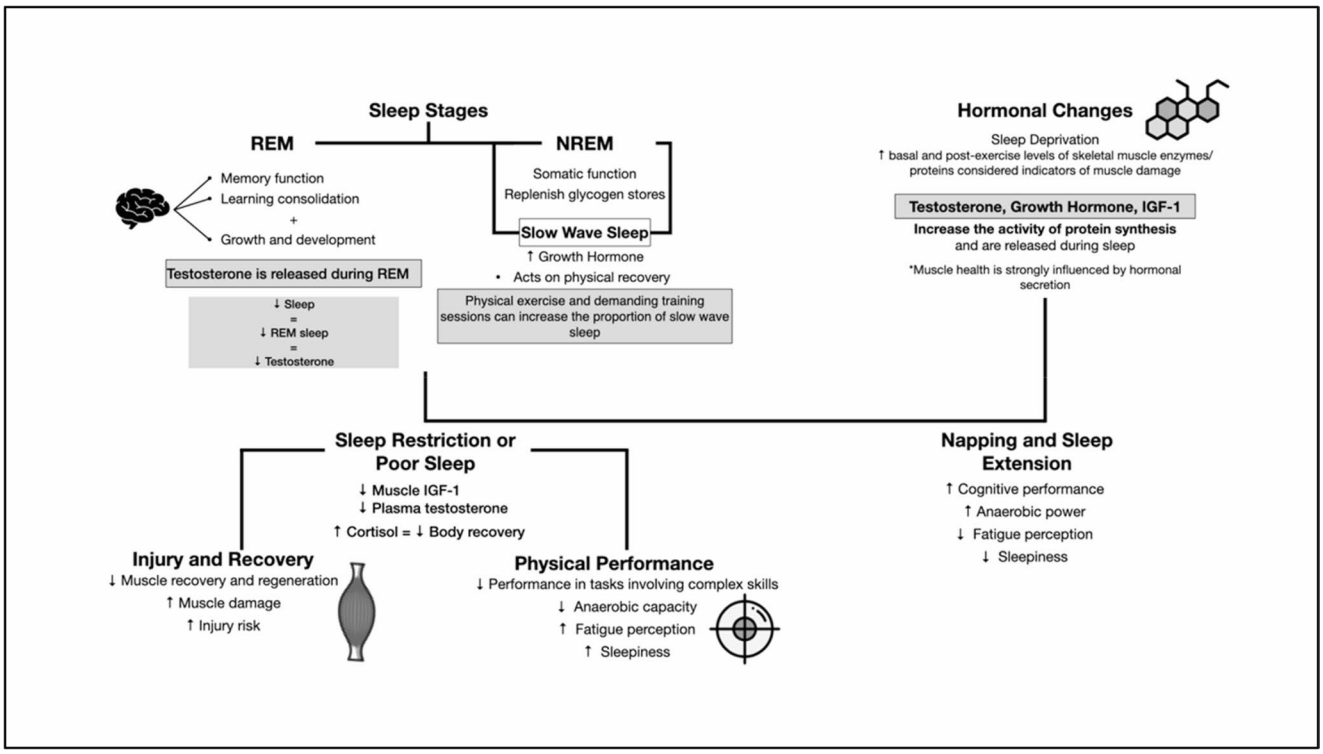

Fig. 1 Schematic of the review's main highlights. ↓ Denotes a decrease; ↑ denotes an increase. Abbreviations: REM, rapid-eye movement; NREM, non-rapid-eye movement; IGF-1, insulin like growth factor 1

(including REM propensity), thereby potentially impairing recovery even if sleep duration is preserved [113, 114].

Importantly, chronotype differences also influence vulnerability to circadian disruption: while evening-types commonly report poorer sleep quality under early-start social schedules and demonstrate greater "social jetlag," morning-types may be comparatively less tolerant to late-night schedules and might struggle to remain awake when circadian sleep propensity is advanced (i.e., when melatonin onset and sleep drive occur early) [115]. Thus, morning-types can also experience substantial sleep disturbance when required to remain awake far past their phase or when rapid time zone changes require prolonged wakefulness before re-entrainment [115, 116]. These effects are relevant for long-haul travel and multi-time-zone competition because the direction and magnitude of the required circadian shift interact with chronotype to determine adaptation speed.

Recommendations

Although evidence points to the benefits of napping and sleep extension, we observed high heterogeneity among study designs, making it difficult to compare findings. In the future, general design recommendations for this type of study would help with pooled results, study replication, and development of new systematic reviews and meta-analyses. This could include baseline period recommendations (i.e., setting, duration, equipment, etc.), protocols for the establishment of sleep extension and restriction protocols (i.e., sleep need, based on average duration), subjects' characteristics to report, and others. Also, despite the advancements and known recommendations of sleep quantity, quality, and others (i.e., sleep regularity), there is still a lack of established biomarkers of sleep. This would potentially aid in an objective assessment of individual sleep health and needs. A summary of this review's findings is presented in Fig. 1.

Conclusions

Multiple processes occur in our brain and body while sleeping, which can affect the quality and timeframe of physical and injury recovery and improve skill learning. SWS plays a pivotal role in physical recovery, with increases in SWS being linked to rises in metabolic stress and growth hormone production, which support anabolic processes. Acute sleep deprivation or restriction may influence sports performance differently, depending on the sport or the modality (i.e., predominantly aerobic or anaerobic). In sports that require executing complex skills or present high cognitive demands, such as tennis, impairments may be more marked as indicated by poorer performance in alertness, reaction time, and

decision-making tests. Nonetheless, research on this topic is scarce and should be explored further. Intervention studies with napping and sleep extension show interesting potential of both strategies, but further research is needed, especially on sleep extension.

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Data availability The manuscript has no associated data.

Declarations

Ethical approval This article does not contain any studies with human participants performed by any of the authors.

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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